

PAPER_672: UQFF Evaporation Timescale

Subtitle: Full UQFF evaporation timescale formula, mass boundary calculations for universe-age crossings, and U_m sensitivity sweep. **Module:** UQFFEvaporationTimescale

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Abstract

We derive the complete UQFF black hole evaporation timescale and calculate the mass at which τ_{UQFF} equals the Hubble time, demarcating the boundary between stable and evaporating black holes.

1. Standard Timescale

$$\tau_{\text{Hawking}} = \frac{5120\pi G^2 M^3}{\hbar c^4}$$

2. UQFF Timescale

$$\tau_{\text{UQFF}} = \frac{\tau_{\text{Hawking}}}{1 - f_{\text{TRZ}}} \cdot \frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}} \cdot \exp\left(\frac{U_m}{k_B T_H}\right)$$

Factor $\approx 30\times (1.11 \times 10 \times 2.718)$.

3. Universe-Age Boundary Masses

Standard: $M_{\text{cross,std}} = \left(\frac{\hbar c^4 t_H}{5120\pi G^2}\right)^{1/3} \approx 5.5 \times 10^{11} \text{ kg}$

UQFF: $M_{\text{cross,UQFF}} = M_{\text{cross,std}} / (30)^{1/3} \approx 1.8 \times 10^{11} \text{ kg}$

UQFF shifts the evaporation boundary $3.1\times$ lower in mass.

4. Sensitivity to U_m

U_m/k_BT_H	τ factor over τ_{std}
0	11.1
1	30.2
2	82
3	222
5	1660

5. Physical Implications

PBHs in the previously evaporating mass range 1.8×10^{11} – 5.5×10^{11} kg become stable under UQFF.

6. C++ Module

UQFFEvaporationTimescale.h / .cpp — Session 172 CP4 #256 — UQFFEvaporationTimescaleCalculator

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